

Machine Learning for Solid-State Battery Materials Discovery: A Survey

Solid-state batteries promise safer, higher-energy electrochemical storage, but their materials space is large, chemically constrained, and difficult to search experimentally. Machine learning can accelerate discovery by predicting ionic conductivity, diffusion barriers, phase and electrochemical stability, interface compatibility, synthesis likelihood, and structure-property relationships for solid electrolytes, electrodes, coatings, and interphases. This survey synthesizes an arXiv-derived corpus of 107 papers collected during this run and organizes the literature around materials informatics, solid electrolyte screening, crystal graph learning, machine-learning interatomic potentials, active learning, Bayesian optimization, generative and foundation models, interface stability, chemistry families, datasets, and experimental validation. The survey argues that useful machine learning for solid-state battery discovery must combine physics-aware representations, uncertainty, high-throughput computation, synthesis constraints, and closed-loop experimental evidence.

Keywords:

solid-state batteries; solid electrolytes; materials discovery; machine learning; ionic conductivity; crystal graph networks; active learning; materials informatics

1 Introduction

Solid-state batteries replace flammable liquid electrolytes with solid ion conductors and promise safer cells, higher energy density, lithium-metal anodes, wider operating windows, and new pack architectures. The central materials problem is demanding: an ideal solid electrolyte should combine high room-temperature ionic conductivity, wide electrochemical stability, chemical and mechanical compatibility with electrodes, processability, low cost, synthesizability, and resistance to dendrites and interfacial degradation. No single descriptor is sufficient, so discovery requires reasoning across structure, chemistry, defects, transport, interfaces, synthesis, and device context [9, 15, 26, 27, 43, 55, 59, 91].

Machine learning enters this problem because the candidate space is enormous. Crystal structures, compositions, substitutions, dopants, grain boundaries, coatings, and processing conditions create a combinatorial search space that cannot be explored by experiment alone. High-throughput density functional theory, molecular dynamics, mate-

rials databases, and experimental reports provide training signals, while graph neural networks, kernel models, transformers, Bayesian optimization, active learning, and machine-learning potentials provide search and prediction tools [6, 15, 16, 43, 46, 55, 59, 91].

The goal is not merely to predict one scalar property. Solid-state battery materials discovery requires a funnel: identify plausible chemistries, estimate stability, predict ion transport, assess electronic insulation, model interfaces, rank synthesizable compounds, prioritize experiments, and update models when measurements disagree with computation. A model that predicts ionic conductivity but ignores electrochemical decomposition or synthesis feasibility may be scientifically interesting yet unusable for discovery. Conversely, a modest model with calibrated uncertainty and an experimental loop can materially accelerate the search [15, 16, 31, 40, 41, 53, 57, 69].

This survey reviews machine learning for solid-state battery materials discovery. We focus on solid electrolytes and related solid-state battery materials, while also covering representation learning, inter-

atomic potentials, high-throughput screening, active learning, inverse design, generative models, interfaces, synthesis, datasets, and validation.

The review takes a materials-discovery perspective rather than a battery-management perspective. We therefore emphasize models that change which compounds, structures, dopants, coatings, or experiments are selected next. Work on cell-level state estimation, charging policies, or lifetime prediction is relevant to battery engineering, but it is outside the scope unless it directly informs materials choice or electrochemical mechanisms. This boundary is important because the data, validation standards, and failure modes differ sharply between materials discovery and operational prediction.

2 Background

2.1 Solid-State Battery Materials

Solid-state battery materials include inorganic solid electrolytes, polymer electrolytes, composite electrolytes, electrode coatings, cathode and anode materials, and interphase-forming compounds. Important electrolyte families include garnets such as LLZO, sulfides such as LGPS and argyrodites, NASICON and LISICON conductors, oxides, halides, perovskites, and solid polymers. Each family trades conductivity, stability, processability, moisture sensitivity, mechanical behavior, and interface compatibility differently [9, 34, 39, 40, 46, 55, 59, 98].

Ion transport depends on crystal topology, mobile ion concentration, bottleneck size, defect chemistry, site disorder, phonons, grain boundaries, and temperature. Stability depends on formation energy, phase equilibria, electronic structure, electrochemical decomposition, and reaction with electrodes. Interfaces introduce additional challenges: space-charge layers, chemical reactions, mechanical stress, dendrite nucleation, and contact loss. Machine learning must therefore represent both local atomic environments and system-level design constraints [9, 15, 26, 27, 43, 55, 59, 91].

The chemistry of solid-state batteries is also conditional on manufacturing. A sulfide that is attractive in a computed bulk phase may fail because of air sensitivity, side reactions, or poor interfacial contact. An oxide with excellent stability may require high-temperature sintering or suffer from grain-boundary resistance. A polymer electrolyte may be processable but temperature dependent. Machine learning studies are most useful when these contextual constraints are made explicit instead of hidden behind

a single target label.

2.2 Materials Informatics

Materials informatics connects databases, descriptors, simulations, and learning algorithms. Classical pipelines use composition descriptors, structural features, handcrafted ion-transport descriptors, and regression or classification models. Modern pipelines increasingly use crystal graphs, equivariant neural networks, transformers, and learned representations from large materials corpora. High-throughput workflows combine first-principles calculations with screening rules to prioritize compounds before synthesis [6, 15, 16, 43, 46, 55, 59, 91].

A recurring distinction is between interpolation and extrapolation. Interpolation within a known family can support rapid ranking of dopants or compositions, while extrapolation to a new chemistry family requires stronger priors and uncertainty estimates. Many optimistic retrospective scores collapse when the split withholds entire structure types, composition regions, or publication years. For discovery, such harder splits are often more informative than random cross-validation.

2.3 Discovery Objectives

Solid-state battery discovery is multi-objective. Desired materials should be conductive for Li or Na ions, electronically insulating, thermodynamically and electrochemically stable, compatible with electrodes, mechanically adequate, processable into dense layers, inexpensive, and synthesizable. Some objectives are cheap to approximate; others require expensive simulations or experiments. This mismatch motivates surrogate models, uncertainty estimates, active learning, and multi-fidelity optimization [15, 16, 31, 40, 41, 53, 57, 69].

Multi-objective discovery also changes the meaning of success. A model with slightly worse mean absolute error can be more valuable if it ranks the top candidates better, avoids unstable regions, or identifies when it is uncertain. The right objective may be expected improvement under constraints, probability of satisfying all filters, Pareto-front diversity, or expected experimental information gain rather than a single-property loss.

3 Methods

3.1 Search Protocol and Corpus Construction

The corpus for this survey was generated during this run from the arXiv structured API. Queries combined solid-state batteries, all-solid-state batteries, solid electrolytes, materials discovery, machine learning, materials informatics, high-throughput screening, ionic conductivity, lithium and sodium conductors, superionic conductors, diffusion and migration barriers, electrochemical and phase stability, interface stability, Bayesian optimization, active learning, DFT, Materials Project, OQMD, AFLOW, crystal graph networks, CGCNN, MEGNet, ALIGNN, M3GNet, machine-learning potentials, garnet, LLZO, argyrodite, LGPS, NASICON, sulfide, oxide, halide and polymer electrolytes, generative models, foundation models, autonomous materials discovery, and synthesis prediction. Records were retained when title or abstract evidence explicitly connected solid-state battery or solid-electrolyte materials with machine learning, high-throughput computation, materials informatics, property prediction, optimization, or experimental discovery. Duplicates were merged by arXiv identifier, and the final bibliography cites 107 unique scholarly preprints.

3.2 Taxonomy

We organize the literature along eight axes. The first is *target material*: sulfide, oxide, halide, garnet, NASICON, LISICON, perovskite, polymer, composite electrolyte, electrode coating, or interface material. The second is *target property*: ionic conductivity, migration barrier, activation energy, diffusion coefficient, phase stability, electrochemical window, elastic response, interface reaction, voltage, capacity, or synthesis likelihood. The third is *data source*: DFT, molecular dynamics, experimental reports, high-throughput databases, literature mining, or autonomous experiments. The fourth is *representation*: composition descriptor, local environment, crystal graph, line graph, equivariant geometry, text, spectra, or interatomic potential. The fifth is *learning task*: regression, classification, ranking, generative design, inverse design, surrogate modeling, or uncertainty estimation. The sixth is *search strategy*: screening, Bayesian optimization, active learning, evolutionary search, reinforcement learning, or closed-loop experimentation. The seventh is *validation*: cross-validation, composition holdout, structure holdout, temporal split, prospective computation, synthesis, or device testing. The eighth is *deployment*: data provenance, reproducibility, interpretability, uncertainty, synthesis constraints, and

integration with electrochemical workflows.

This taxonomy is intentionally workflow-oriented. In practice, a discovery pipeline may begin with generative or enumerative candidate construction, apply thermodynamic and electrochemical filters, predict transport with a surrogate or machine-learning potential, estimate synthesis feasibility, and then allocate a small number of experiments. Papers that appear methodologically different can therefore occupy adjacent roles in the same funnel.

3.3 Materials Informatics and High-Throughput Screening

High-throughput screening is often the first stage of discovery. Candidate structures are enumerated or drawn from databases, then filtered by stability, band gap, ion migration, conductivity proxies, elemental abundance, and interface compatibility. Machine learning can replace expensive calculations, correct low-fidelity predictions, or prioritize which DFT and molecular-dynamics jobs to run. The value of these models depends on how well the screening criteria match device-level requirements [6, 15, 16, 43, 46, 55, 59, 91].

Screening studies should therefore report the assumptions behind each filter. A stability cutoff based on a computed convex hull, a conductivity threshold inferred from a migration barrier, or an electrochemical window predicted from decomposition energies all encode approximations. Transparent reporting lets later work decide whether a candidate was rejected because it is chemically implausible or because the proxy was too conservative.

3.4 Solid Electrolyte Property Prediction

Solid electrolytes are the core target for many studies. Models predict ionic conductivity, activation energy, diffusion barriers, mobile-ion pathways, site disorder, and transport dimensionality. Some use explicit structural descriptors; others learn from crystal graphs or molecular-dynamics trajectories. Because transport can depend on defects, disorder, temperature, grain boundaries, and synthesis history, property labels are heterogeneous. Robust models should report uncertainty and distinguish intrinsic crystal transport from practical electrolyte performance [9, 15, 26, 27, 43, 55, 59, 91].

A particular challenge is that fast ion transport is often a rare-event problem. Short molecular-dynamics trajectories may not observe enough hops, while

static structure descriptors may miss correlated motion and disorder. Learning from barriers, pathways, local coordination, vibrational features, and long-timescale surrogate dynamics can provide complementary views, but the resulting labels should not be treated as interchangeable.

3.5 Crystal Graph Learning and Machine-Learning Potentials

Crystal graph neural networks and related architectures encode atoms, bonds, periodic neighborhoods, and sometimes line graphs or equivariant geometry. They support prediction of formation energy, band gaps, elastic properties, migration barriers, and transport descriptors. Machine-learning interatomic potentials extend this idea to dynamics: they can accelerate molecular dynamics and sample diffusion pathways that would be too expensive with first-principles methods. Their usefulness depends on training-set coverage and failure detection under new chemistries [8, 9, 15, 16, 43, 46, 55, 59].

For solid electrolytes, machine-learning potentials should be evaluated against the quantities that matter for transport, not only force error. A potential with low average force error can still distort a diffusion pathway, stabilize an incorrect local environment, or miss a transition state. Benchmarks that compare conductivity, activation energy, radial distribution functions, site occupancies, and failure cases are more informative than a single aggregate metric.

3.6 Active Learning, Bayesian Optimization, and Inverse Design

Active learning and Bayesian optimization are natural for battery materials discovery because labels are costly. A model selects candidates expected to improve the search, either by high predicted performance, high uncertainty, or a balance of exploitation and exploration. Inverse design can search composition and structure space for materials satisfying multiple constraints. The hard part is defining acquisition functions that reflect the actual discovery objective: conductivity alone is insufficient without stability, synthesizability, and interface compatibility [15, 16, 31, 40, 41, 53, 57, 69].

Closed-loop optimization further requires practical constraints on batch size, measurement time, failed syntheses, and changing laboratory priorities. The next best computation is not always the next best experiment. A useful optimizer must handle asynchronous feedback, multi-fidelity labels, and the pos-

sibility that a failed synthesis is itself an informative negative result.

3.7 Generative and Foundation Models

Generative models propose compositions, crystal structures, substitutions, dopants, or processing recipes. Foundation models and language models can transfer representations across materials tasks, use text or database metadata, and support literature-guided discovery. These methods are attractive when paired data are sparse, but they require strong validity checks. A generated crystal must obey chemistry, charge balance, symmetry, stability, and synthesis constraints before it becomes a credible battery candidate [15, 16, 22, 31, 38, 43, 53, 81].

3.8 Interfaces, Stability, and Electrochemical Performance

Solid-state battery performance is often limited by interfaces rather than bulk electrolyte conductivity. Machine learning can help predict electrochemical stability windows, decomposition products, cathode-electrolyte reactions, lithium-metal compatibility, coating effectiveness, and mechanical failure. Interface datasets are smaller and more context-dependent than bulk datasets, so physically informed descriptors and multi-fidelity data are important [9, 15, 26, 27, 43, 46, 55, 91].

Interface modeling also exposes a scale mismatch. Atomic calculations can identify reaction products and local decomposition mechanisms, but device performance depends on morphology, stack pressure, contact area, current density, and cycling history. Machine learning can bridge some of these scales only if models are explicit about which variables are observed, which are simulated, and which are assumed.

3.9 Chemistry Families

Different material families pose different modeling problems. Sulfides can offer high conductivity but may be moisture sensitive and chemically reactive. Oxides and garnets can be more stable but challenging to densify and interface with lithium metal. Halides are emerging for cathode compatibility. NASICON and LISICON structures provide tunable conduction frameworks. Polymer electrolytes add segmental dynamics and processing variables. Machine learning should exploit these family-specific priors while still enabling cross-family transfer [9, 34, 39, 40, 46, 55, 59, 98].

3.10 Datasets, Synthesis, and Validation

Data quality is a limiting factor. Computed databases are large but may omit disorder, defects, interfaces, and synthesis history. Experimental datasets are smaller, noisier, and biased toward publishable chemistries. Validation should include composition or chemistry holdouts, prospective screening, uncertainty calibration, and where possible synthesis and electrochemical testing. Benchmarks should report not only prediction error but also discovery utility: how many high-value candidates are found per computation or experiment [6, 9, 15, 16, 26, 43, 46, 91].

Reproducibility requires more than publishing a trained model. Discovery claims should include candidate lists, failed filters, data versions, relaxation settings, feature definitions, split protocols, uncertainty calibration, and synthesis or measurement metadata when available. Without these details, it is difficult to determine whether a new electrolyte was found by the model, inherited from the dataset, or created by an unreported heuristic.

3.11 Representative Literature Map

The following map cites the retained corpus by theme, making the coverage explicit across materials informatics, solid-electrolyte transport, graph learning, machine-learning potentials, active learning, generative models, interface stability, chemistry families, datasets, synthesis, and validation.

Materials informatics and high-throughput screening The materials informatics and high-throughput screening cluster contains work that develops methods, systems, datasets, or analyses central to this survey [6, 8, 16, 32, 39, 46, 55, 59, 107]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [7, 12, 18, 30, 31, 78, 79, 84, 89]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [1, 4, 19, 37, 44, 52, 61, 95, 106]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [90].

Solid electrolytes and ion transport The solid electrolytes and ion transport cluster contains work that develops methods, systems, datasets, or analyses central to this survey [9, 26, 27, 34, 38, 43, 57, 76, 98]. A second strand in this cluster studies de-

sign trade-offs, evaluation protocols, failure modes, and domain-specific variants [3, 33, 35, 40, 66, 69, 80, 88, 97]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [5, 14, 20, 42, 62, 70, 73, 93, 102]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [11, 17, 25, 45, 49, 51, 58, 86, 87]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [23, 75].

Crystal graph learning and machine-learning potentials The crystal graph learning and machine-learning potentials cluster contains work that develops methods, systems, datasets, or analyses central to this survey [22, 56, 65, 72, 92, 94, 99, 103].

Active learning, bayesian optimization, and inverse design The active learning, Bayesian optimization, and inverse design cluster contains work that develops methods, systems, datasets, or analyses central to this survey [15, 21, 36, 41, 47, 50, 53, 67, 101].

Generative and foundation models for battery materials The generative and foundation models for battery materials cluster contains work that develops methods, systems, datasets, or analyses central to this survey [28].

Stability, interfaces, and electrochemical performance The stability, interfaces, and electrochemical performance cluster contains work that develops methods, systems, datasets, or analyses central to this survey [2, 10, 13, 63, 68, 77, 82, 83, 96]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [29, 71, 104].

Datasets, validation, synthesis, and deployment The datasets, validation, synthesis, and deployment cluster contains work that develops methods, systems, datasets, or analyses central to this survey [24, 48, 54, 60, 64, 81, 85, 91, 105]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [74, 100].

4 Challenges and Prospects

4.1 Sparse, Biased, and Multi-Fidelity Data

Solid-state battery datasets mix computed and experimental labels with different errors and biases. Future models should represent provenance, uncertainty, temperature, synthesis history, and fidelity rather than treating every label as equally reliable.

A promising direction is hierarchical data modeling. First-principles calculations, molecular dynamics, literature measurements, and laboratory data can be treated as related but distinct views of the same underlying property. Such models can learn systematic offsets between fidelities and decide when an expensive label is worth acquiring.

4.2 From Bulk Properties to Device-Relevant Criteria

High ionic conductivity is necessary but not sufficient. Discovery workflows need multi-objective screening for stability, interface compatibility, electronic insulation, mechanics, processability, and cost. This requires models that compose property predictions into decision rules.

The field would benefit from benchmark tasks that explicitly score constrained discovery. Instead of asking for the best conductivity predictor, a benchmark could ask which ten candidates should be synthesized under constraints on stability, elemental abundance, and interface compatibility. This would align evaluation with actual materials decisions.

4.3 Disorder, Defects, and Interfaces

Many high-performing conductors depend on disorder, defects, dopants, and interfaces. These are hard to encode from ideal crystal structures alone. Promising directions include generative defect models, graph representations of disordered sites, and machine-learning potentials for realistic microstructures.

4.4 Closed-Loop Discovery

The most useful systems will connect prediction to action: select candidates, run simulations or syntheses, measure properties, update the model, and document failures. Active learning and autonomous laboratories should be evaluated by their ability to discover materials, not only by retrospective error metrics.

Closed-loop systems also need negative knowledge. Failed syntheses, unstable phases, poor contacts, and

low-conductivity measurements are often underreported, yet they are exactly the data that prevent models from repeatedly suggesting unworkable regions of composition space.

4.5 Interpretability and Scientific Insight

Battery materials discovery needs explanations that materials scientists can use. Models should expose which structural motifs, bottlenecks, substitutions, interfaces, or synthesis factors drive predictions, and should separate known chemical priors from learned correlations.

5 Conclusion

Machine learning is reshaping solid-state battery materials discovery by linking materials databases, high-throughput computation, crystal graph learning, interatomic potentials, active learning, generative design, and experimental validation. The field’s central challenge is integration: models must reason across conductivity, stability, interfaces, synthesis, uncertainty, and device constraints. Progress will come from physics-aware representations, calibrated multi-fidelity learning, prospective benchmarks, and closed-loop workflows that turn predictions into verified materials.

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